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Milestone 1: ML Solution EDA

Learners Name	<i>Aira Joie Piopongco</i>
Program and Specialization	<i>BSIT - Data Analytics</i>

I. Introduction

The dataset is from FinMark Corporation, which provides cloud-based finance operations platforms, from payroll, accounting, tax computation, and regulatory reporting to small and medium enterprises (SMEs) across Southeast Asia. A Customer Demographics dataset describing who FinMark's SME account holders are (*age, gender, location, income level, and signup date*), a Customer Transactions dataset describing platform usage and payment activity (*transaction amount, product category, payment method, and transaction date*), and a Social Media Interactions dataset describing how customers engage with the FinMark brand online (*platform, interaction type, and sentiment*). Together, these three files were designed to provide FinMark with a rounded picture of its customers' relationship.

The datasets are intended to support business and machine learning activities involving customer analysis. The combination of demographic, transaction, and social media information can potentially be used in future activities for customer profiling, behavioral analysis, segmentation, prediction, and other machine learning applications. The business struggles to derive actionable insights from them because of the overwhelming volume of data and inconsistencies across sources, which limits its ability to identify market trends, understand customer behavior, and support data-driven decision-making and strategic planning. The main objective of this is to understand the structure and characteristics of the three datasets, identify data-quality problems, and document the preprocessing procedures, developing a solution that turns this inconsistent data into actionable insights for the intended audience, FinMark, offering them a solution that aligns with what the company intends to deliver.

Doing the dataset preprocessing and exploratory data analysis is important before developing solutions because raw datasets may contain inconsistent values that can affect the statistical analysis and may negatively affect the performance and reliability of future machine learning models. In the preprocessing activities, duplicate records were removed, missing values were handled, invalid numerical values were corrected, and date variables were standardized. Exploratory data analysis was then performed to examine numerical distributions and categorical frequencies. The preprocessing and completion of EDA establish a cleaner and more consistent foundation for future machine learning tasks by ensuring that variables have appropriate formats, missing values are addressed, duplicate records are removed, and important characteristics of the datasets are documented.

II. Dataset Overview

The three datasets are used to cover each differing facet of the company's customer relationship. The raw datasets were contaminated CSV files containing 3,200 records and 6 columns, which called for data cleaning. After cleaning, each dataset retains between 3,015 and 3,023 unique records, with row counts differing slightly across datasets because each file underwent an independent deduplication and validation process. The three datasets share a common CustomerID key, which means they can eventually be joined into a single customer-level view for modeling, even though they are analyzed separately in this report.

The datasets contain a mixture of numerical and categorical variables, as well as date variables. Customer Demographics contains Age as a numerical variable and Gender, Location, and IncomeLevel as categorical variables. Customer Transactions contains Amount as a numerical variable and ProductCategory and PaymentMethod as categorical variables. Social Media Interactions consists primarily of categorical variables, with InteractionDate representing the date of the interaction. CustomerID, TransactionID, and InteractionID serve primarily as identifiers.

A. Dataset Structure

Dataset	No. of Rows (after cleaning)	Columns	Main Variables and their data types (after preprocessing)
Customer Demographics	3,023	6	- CustomerID (object) - Age (float64) - Gender (object) - Location (object) - IncomeLevel (object) - SignupDate (datetime)
Customer Transactions	3,015	6	- CustomerID (object) - TransactionID (object) - TransactionDate (datetime) - Amount (float64) - ProductCategory (object) - PaymentMethod (object)
Social Media Interactions	3,020	6	- CustomerID (object) - InteractionID (object) - InteractionDate (datetime) - Platform (object) - InteractionType (object) - Sentiment (object)

- Customer Demographics - The dataset is organized at the customer-record level, although CustomerID values may occur more than once in the original data. The preprocessing workflow removed complete duplicate rows, resulting in 3,023 remaining records.
- Customer Transactions - The dataset represents customer transaction records. The transaction dates cover the period from July 1, 2022, to June 30, 2024. The transaction amount is a numerical variable after preprocessing.

- Social Media Interactions - The interaction dates cover the period from July 1, 2023 to June 30, 2024. The main categorical variables describe the platform used, type of interaction, and sentiment.

B. Initial Dataset Observations

- Several data-quality issues were identified before preprocessing. All three original datasets contained duplicate rows.
- Missing values were also present.
- The datasets also contained inconsistent or invalid values. In Customer Demographics, Age was initially stored as an object rather than a numerical variable. 30 occurrences of an age value of -1, and 33 occurrences of an age value of 150.
- In Customer Transactions, Amount was initially stored as an object and contained "Free" values and -100 values.
- Transaction dates also appeared in two formats: YYYY-MM-DD and DD/MM/YYYY. Similar mixed date formats were identified in SignupDate and InteractionDate.
- Some variables may become important. Age, IncomeLevel, transaction Amount, ProductCategory, PaymentMethod, Platform, InteractionType, and Sentiment may provide useful behavioral or customer-related features.

III. Data Preprocessing Summary

Area	Tasks
Duplicate Rows	Every acquired dataset had duplicates. Detected duplicate records using <code>df.duplicated()</code> and removed them using <code>df.drop_duplicates()</code> . Removing duplicate records was necessary because repeated observations could give disproportionate importance to certain customers or

	transactions and could distort frequency counts and descriptive statistics The Customer Demographics dataset contained 177 duplicate rows, the Customer Transactions dataset contained 185 duplicate rows, and the Social Media Interactions dataset contained 180 duplicate rows. The duplicate records were removed using <code>df.drop_duplicates()</code>. After removal, the Customer Demographics dataset contained 3,023 records, the Customer Transactions dataset contained 3,015 records, and the Social Media Interactions dataset contained 3,020 records. A final <code>df.duplicated().sum()</code> check confirmed that the cleaned datasets contained zero duplicate rows.
Missing Values	Identified missing values using <code>df.isnull().sum()</code>. Numerical and categorical missing values were handled using median imputation for applicable numerical values or "Undetermined" for categories, as appropriate. Ensured missing values were handled according to the type and meaning of each variable. In Customer Demographics, Age initially contained 276 missing values after duplicate removal. Additional invalid Age values were identified, including Unknown, -1, and 150. After converting Age to a numerical type and identifying invalid values, the invalid ages were treated as missing. The resulting missing Age values were replaced using the median age of 45. This resulted in a valid Age range of 18 to 70. IncomeLevel contained 285 missing values. Rather than assigning an assumed income category, these records were replaced with the categorical value "Undetermined". This preserves the fact that the original income



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	information was unavailable. In Customer Transactions, Amount contained 283 missing values after duplicate removal. ProductCategory contained 282 missing values. ProductCategory missing values were replaced with "Undetermined". For Amount, the value "Free" was converted to zero, while -100 values were considered invalid. After conversion and correction, missing transaction amounts were replaced using the median transaction amount of 498.015. In Social Media Interactions, Platform contained 291 missing values and Sentiment contained 309 missing values. Both were replaced with "Undetermined" rather than assigning an arbitrary platform or sentiment. The approaches were selected according to the variable types. Median imputation was used for numerical variables because it provides a central value while being less affected by extreme observations than the mean. For categorical variables where the true category could not be determined, "Undetermined" was used to retain the records without assuming information.
Data Validation and Corrections	Corrected numerical and date data types, identified invalid values in several categories, and standardized date formats. Several invalid and inconsistent data types were identified during preprocessing. In the original Customer Demographics dataset, Age was stored as an object even though it represents a numerical measurement. It was converted to float64 using <code>pd.to_numeric()</code>. The Amount variable in Customer Transactions was also initially stored as an object because it contained both



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	numerical values and text such as "Free". "Free" was converted to zero, after which the column was converted to a numerical float64 type. Invalid negative -100 values were then treated as missing and replaced with the median. The date variables were initially stored as objects. SignupDate, TransactionDate, and InteractionDate contained dates using both YYYY-MM-DD and DD/MM/YYYY formats. The preprocessing workflow parsed the first format and then handled the alternative format using <code>pd.to_datetime()</code>. The resulting variables were converted to <code>datetime64[ns]</code>. Numerical variables must be recognized as numerical values for statistical calculations, while date variables need to be recognized as dates before features such as year, month, day, or time intervals can be extracted. Correct data types also reduce the possibility of errors during future analysis.
Additional Preprocessing Procedures	Additional cleaning procedures were performed to improve consistency. In the Customer Demographics dataset, invalid ages below zero and above 100 were removed from consideration and subsequently imputed. The unknown age values were also treated as invalid numerical entries and labeled as unknown.. In Customer Transactions, "Free" was interpreted as a zero transaction amount, while the repeated -100 values were treated as invalid. ProductCategory missing values were represented using "Undetermined". For Social Media Interactions, missing Platform and Sentiment values were represented using "Undetermined". The InteractionDate variable was

	standardized from both date formats into a datetime variable. No variable renaming or major restructuring was performed in the preprocessing workflows. The original variable names were retained so that the cleaned datasets remained consistent with the source datasets.
Preprocessing Challenges and Observations	One of the main preprocessing challenges was that several variables appeared numeric or date-based conceptually but were stored as object variables. This required additional validation and conversion before statistical analysis could be performed. Another challenge was distinguishing genuinely missing information from invalid information. For example, missing income level and missing social media platform could reasonably be represented as "Undetermined", whereas an age of -1, an age of 150, and a transaction amount of -100 were treated as invalid observations. The transaction amount value "Free" required a separate interpretation and was converted to zero. After preprocessing, the datasets showed significant improvements in consistency. All three final datasets had zero duplicate rows and zero missing values. Numerical variables were represented using numerical data types, and date variables were standardized to datetime format during preprocessing. These improvements provide a cleaner foundation for the subsequent EDA.

IV. Exploratory Data Analysis

A. Descriptive Statistics

The main numerical variables examined were Customer Age and Transaction Amount.

Age (Customer Demographics) has a mean of 44.69, a median of 45.00, and a mode of 45.00, indicating a roughly symmetric, mildly bell-shaped distribution rather than a heavily skewed one. Ages span the full plausible adult range from 18 to 70 (a range of 52 years) with a standard deviation of 14.42 and a variance of 208.05, meaning most customers fall within roughly one standard deviation (about 30 to 59 years old) of the average. Observation of a moderate, not extreme, spread. This is the expected result of having removed the -12 and 150 sentinel values during preprocessing, which would otherwise have artificially widened the range and inflated the variance.

Amount (Customer Transactions) has a mean of 493.62 and a median of 498.02, which are close enough to each other to suggest the bulk of transaction values are fairly evenly spread rather than concentrated at one extreme. The mode also lands on 498.02, reflecting the median-imputed values used to fill missing entries. Its standard deviation (279.30) is large relative to its mean, roughly 57% of the mean, indicating substantial variability in how much is transacted per record, though the variance of 78,009.07 is proportionate to a variable spanning nearly a 1,000-unit range (0.00 to 999.86) and does not by itself indicate a remaining data problem, since this figure already reflects the cleaned data. Sixty transactions (about 2.0% of the cleaned dataset) carry an Amount of exactly 0, the majority of which trace back to the "Free" transactions identified during verification, these were intentionally preserved as legitimate zero-value events rather than dropped or treated as ordinary purchases.

B. Frequency and Distribution Analysis

The categorical variables generally show relatively balanced distributions, although several datasets contain an "Undetermined" category resulting from missing information.

Gender is close to an even split, with 1,528 customers (50.55%) being Female and 1,495 (49.45%) being Male, showing no meaningful imbalance.

IncomeLevel is fairly even across its three substantive categories, High (946 customers, 31.29%), Low (923, 30.53%), and Medium (869, 28.75%) with the remaining 285 customers (9.43%) carrying the “Undetermined” label assigned during missing-value imputation, because this is a real, non-trivial share of the data, it should be treated as a genuine fourth category in any downstream segmentation model rather than dropped, since dropping it would silently discard nearly 1 in 10 customers.

ProductCategory is likewise close to evenly distributed across its five substantive groups: Clothing (571, 18.94%), Electronics (564, 18.71%), Automotive (540, 17.91%), Home & Garden (530, 17.58%), and Health & Beauty (528, 17.51%) — with 282 transactions (9.35%) labeled “Undetermined.” No single category dominates purchase behavior, which will make category-based segmentation genuinely informative rather than trivial. PaymentMethod is even more tightly balanced, split almost exactly into quarters across Debit Card (767, 25.44%), Credit Card (755, 25.04%), PayPal (750, 24.88%), and Bank Transfer (743, 24.64%), indicating customers show no strong collective preference for any one payment channel.

Platform and InteractionType follow the same near-even pattern: Instagram (921, 30.50%), Twitter (917, 30.36%), and Facebook (891, 29.50%) each account for roughly 30% of interactions, with 291 interactions (9.64%) labeled “Undetermined,” and Comment (1,018, 33.71%), Share (1,007, 33.34%), and Like (995, 32.95%) each account for roughly a third of interaction types. Sentiment is the one variable in the collection with a clearly imbalanced category: the moderate labels dominate, Positive (905, 29.97%), Neutral (874, 28.94%), and Negative (866, 28.68%) while the extreme labels Very Negative (34, 1.13%) and Very Positive (32, 1.06%) are rare by comparison, together making up only about 2.2% of all interactions with a further 309 records, 10.23%, labeled “Undetermined”. This imbalance matters because a future sentiment classification model trained naively on this data would struggle to

learn the extreme classes without resampling or class-weighting, simply due to how few examples exist.

C. Pattern and Trend Analysis

- Several cross-cutting patterns emerge from comparing the three datasets side by side. First, the level of category balance is remarkably consistent across the collection: Gender, IncomeLevel, ProductCategory, PaymentMethod, Platform, and InteractionType are all close to evenly distributed among their substantive categories. This is a favorable characteristic for machine learning, since heavily imbalanced categorical predictors often require special handling (oversampling, class weights, or merging of rare categories), and most of these variables will not need it. Sentiment is the notable exception, and any future model that predicts or uses sentiment should account for the scarcity of its extreme classes.
- The “Undetermined” category introduced during missing-value imputation appears at a strikingly similar rate across independent variables and datasets: 9.43% for Income Level, 9.35% for Product Category, 9.64% for Platform, and 10.23% for Sentiment. This consistency suggests the missing values were injected at a comparable rate across the raw files rather than being concentrated in one problematic column, and it means any model consuming these features should expect roughly 1 in 10 records to carry an “Undetermined” label for any given categorical field.
- Examining customer identifiers reveals repeat activity. Customer Demographics dataset contains 3,023 rows built from 3,000 unique CustomerIDs, while Customer Transactions compresses 3,015 transactions into only 1,871 unique customers, and Social Media Interactions compresses 3,020 interactions into 1,893 unique customers. In other words, the typical FinMark customer in this sample transacts roughly 1.6 times and posts or engages roughly 1.6 times, meaning repeat behavior rather than one-off activity is the norm, which is

encouraging for any future customer lifetime value or customer churn model that depends on having multiple observations per customer.

- The date ranges covered by each dataset differ significantly. SignupDate spans nearly five years (July 2019 to June 2024), TransactionDate spans two years (July 2022 to June 2024), and InteractionDate spans exactly one year (July 2023 to June 2024). Social media activity is therefore only observable for the most recent year of the overall timeline, so any future analysis relating sentiment to earlier transaction or signup behavior will only have a partial, one-year overlapping window to work with. A scope limitation that should be made explicit in later modeling rather than assumed away.
- No strong outliers remain in either numerical variable after cleaning. Age is bounded and moderately spread (18–70, std. dev. 14.42) and Amount is bounded and moderately spread (0–999.86, std. dev. 279.30). The one anomaly worth monitoring going forward is the cluster of 60 zero-value transactions, which are legitimate but should be explicitly modeled like for example, with a separate “is free transaction” flag rather than treated as ordinary purchase amounts in any future revenue-prediction task.

V. Initial Findings and Insights

- The diagnosis of the pre-cleaning data confirms FinMark's own problem statement, the three datasets were individually inconsistent (177–185 duplicate rows each), incomplete (276–309 missing values per affected column), and inappropriate types (numeric and date columns stored as unreliable text), which is precisely the kind of cross-source inconsistency that was blocking actionable insight. After preprocessing, all three datasets are duplicate-free, fully typed, and 100% complete, which resolves that diagnosis at the individual-dataset level. The demographic and transactional variables are demographically and behaviorally balanced (Gender, Income Level, Product Category, and Payment Method are all close to evenly distributed), while Sentiment is the one variable that remains structurally imbalanced even after cleaning, since imputation

cannot manufacture examples of rare sentiment classes that were never collected.

- These findings forecast specific requirements for the modeling milestones that follow. A customer segmentation model built on demographics and spend behavior can be developed with high confidence, since its input variables are balanced, bounded, and free of remaining outliers. A sentiment-driven model, by contrast, will need class-imbalance handling before it can reliably learn the Very Negative and Very Positive classes, and any model that relates sentiment to transaction or signup history will need to explicitly restrict itself to the one-year window in which social data exists, rather than assuming coverage across the full five-year signup history. The 9–10% “Undetermined” share present in every imputed categorical field should also be expected to persist as a real, non-trivial category in any feature built from those columns, rather than as noise that model tuning will eliminate.
- Recommended actions include applying class-imbalance handling specifically to Sentiment before it is used as a predictive feature or target, given the scarcity of its Very Negative and Very Positive classes. Explicitly restricting any time-based analysis that combines Social Media Interactions with the other two datasets to the shared one-year overlap window (July 2023 to June 2024), to avoid drawing conclusions from periods where social data simply does not exist. Revisiting the “Undetermined” categories once additional data sources are available, to decide whether they should remain a standalone category or be re-imputed using a more informed model-based method later on. And lastly, is to treat the 60 zero-value (“Free”) transactions as a distinct, flagged event type in any spend-prediction model, rather than folding them into the ordinary Amount distribution.

VI. Project Resources and Supporting Files

Reference No.	Reference Description	Link
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1	Original Raw Dataset	 FinMark CSV
2	Final Cleaned Dataset	 Cleaned Data... <u>github repository</u>
3	Python Code for Dataset Preprocessing	<u>github repository</u>